

Intro to Transfer Functions

EGR 345

GVSU

Outline

- review
- intro to transfer functions
 - what is a transfer function?
 - how do we find transfer functions?
 - why are they so powerful?
 - what is an impulse response?
 - what is a step response?

Review

- what is a dynamic system?

→ transient response matters
↓
diff. eqn.

Review

- what is a dynamic system?
- why is Dr. Krauss obsessed with mass/spring/damper systems?

→ helps us visualize dynamic systems

→ they exist in many places/things

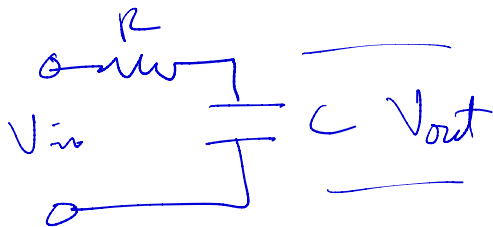
Review

- what is a dynamic system?
- why is Dr. Krauss obsessed with mass/spring/damper systems?
- where can we find MSD systems in real-life?

- vehicle suspensions
- bicycles
- buildings (earthquakes)

What is a transfer function?

- an input/output model that encapsulates the underlying diff. eqn.



$$G(s) = \frac{V_{out}(s)}{V_{in}(s)}$$

assume $y(s) \rightarrow 0$

Find the TF

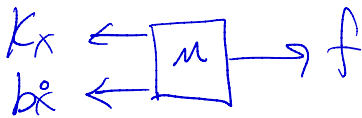
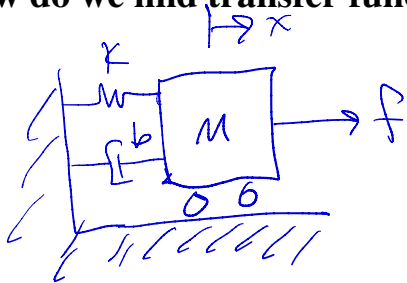
$$\mathcal{L}[y + py = u] = sY(s) + pY(s) = U(s)$$

$$Y(s)(s+p) = U(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s+p}$$

How do we find transfer functions?

input: f output: x



$$m\ddot{x} = -kx - b\dot{x} + f$$

$$\mathcal{L} [m\ddot{x} + b\dot{x} + kx = f]$$

$$G(s) = \frac{X(s)}{F(s)}$$

$$G(s) = \frac{X(s)}{F(s)}$$

$$\mathcal{L} [m\ddot{x} + b\dot{x} + kx = f]$$

$$ms^2 X(s) + bs X(s) + kX(s) = F(s)$$

$$(ms^2 + bs + k) X(s) = F(s)$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} \cdot \frac{1/n}{1/n} = \frac{1/n}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

$$s^2 + \frac{b}{m}s + \frac{k}{m}$$

$$s = \frac{-\left(\frac{b}{m}\right) \pm \sqrt{\frac{b^2}{m^2} - 4\frac{k}{m}}}{2}$$

$$\frac{b^2}{m^2} - 4\frac{k}{m} = 0$$

bcr cause term
under radical to
go to zero

$$b_{cr}^2 = 4km$$

$$b_{cr} = 2\sqrt{km}$$

$$G(s) = \frac{1/m}{s^2 + \frac{b}{m}s + \frac{k}{m}} = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\zeta = \frac{b}{b_{CR}}$$

$$b_{CR} = 2\sqrt{km}$$

$$\frac{b}{m} = 2\zeta\omega_n$$

↑
standard MSD
TF

Always assume zero initial conditions when finding transfer functions.

Yes, always!

What can we do with transfer functions?

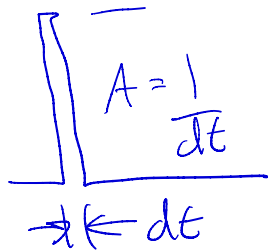


→ find the response of a system to various inputs

- impulse
- step
- sinusoidal

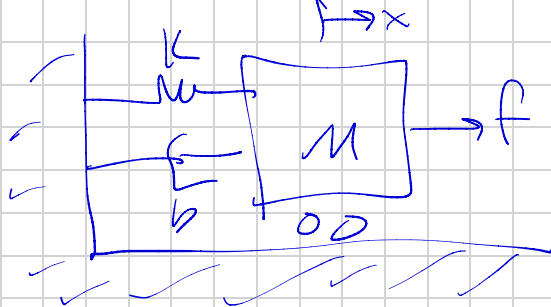
What is an impulse response?

- and how do we find it?
 - “by hand”
 - with python



— large amplitude
for a short
duration

mechanically: whacked with
hammer



$$m = 1 \text{ kg}$$

$$k = 10 \text{ N/m}$$

$$b = 0.5 \frac{\text{N.s}}{\text{m}}$$

$$\mathcal{L}[f] = 1$$

for a unit impulse

$$G(s) = \frac{X(s)}{F(s)} = \frac{Y_M}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Find the unit impulse response
 $\rightarrow x(t)$

Output as function of time

$$G(s) = \frac{X(s)}{F(s)} \rightarrow X(s) = G(s) \cdot \underbrace{F(s)}$$

$$X(s) = \frac{Y_m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\zeta = \frac{0.5}{b_{cr}} = 0.08$$

$$\omega_n = \sqrt{\frac{10}{1}} = 3.16$$

$$b_{cr} = 2\sqrt{km} =$$

$$b_{cr} = 2\sqrt{10} = 6.32$$

$$X(s) = \frac{1}{s^2 + 0.5s + 10} = \frac{A\omega}{(s+a)^2 + \omega^2}$$

$$\underline{s^2 + 0.5s + 10} = \underline{s^2} + \underline{2as} + \underline{a^2 + \omega^2}$$

ω_n^2
 $\downarrow$
 $\uparrow$
 $\frac{k}{m} = \omega_n$

$$0.5 = 2a$$

$$a = 0.25$$

$$\omega = \sqrt{10 - 0.25^2} \equiv \omega_d$$

$$\text{Is } a \neq 0, \omega \neq \omega_n$$

$$A = \frac{1}{\omega}$$

$$\omega_n^2 = a^2 + \omega^2$$

$$\omega^2 = \omega_n^2 - a^2$$

$$X(s) = \frac{A \omega}{(s+a)^2 + \omega^2}$$

See Python
example
notebook

$$x(t) = A e^{-at} \sin(\omega t)$$

$$x(t) = 0.317 e^{-0.25t} \sin(3.15t)$$

$$\frac{1}{s^2 + \frac{b}{n}s + \frac{k}{m}} = \frac{A \omega}{(s+a)^2 + \omega^2} \leftarrow \text{Row 8 from Laplace Table}$$

$$\frac{1}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{A \omega}{(s+a)^2 + \omega^2}$$

$$\underline{s^2 + \frac{b}{n}s + \frac{k}{m}} = \underline{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\zeta^2 \epsilon \sum \omega_n \zeta \omega_n^2 = \zeta^2 \epsilon \sum \omega_n + \zeta^2 \epsilon \omega^2$$

$$\omega_n^2 = \omega^2 + a^2 \epsilon \omega^2$$

What is a step response?

- and how do we find it?
 - “by hand”
 - with python

What is a step function?

- how is $1(t)$ different from just the number 1?